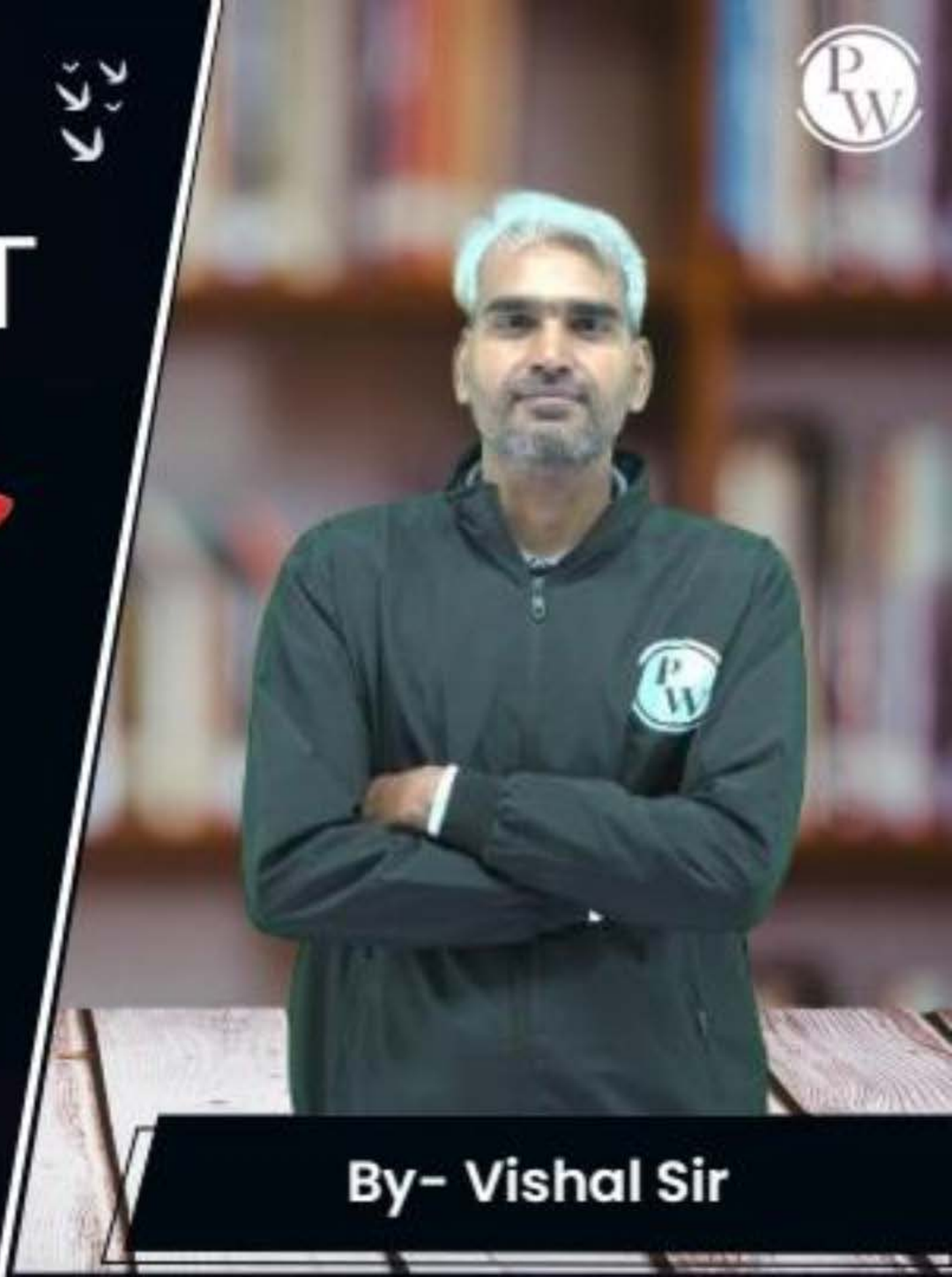
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic Schedule

Topic ACID properties

Slide

Topics to be Covered



- Topic: Equivalent schedule
- Topic: Serializable schedule
- Topic: RW problem
- Topic: WR problem
- Topic: WW problem
- Topic: Lost update problem

Slide



Topic : Equivalent schedule

For two schedules to be equivalent.

(i) Every read must be same in both schedules

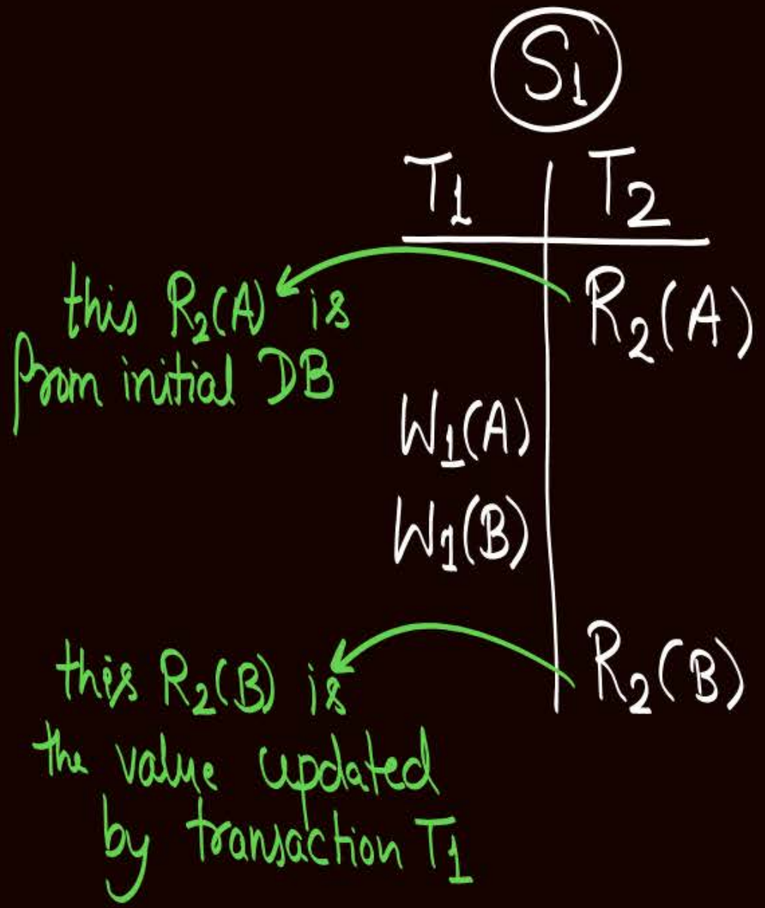
(ii) Final update of every data item should also be same in both the schedules.

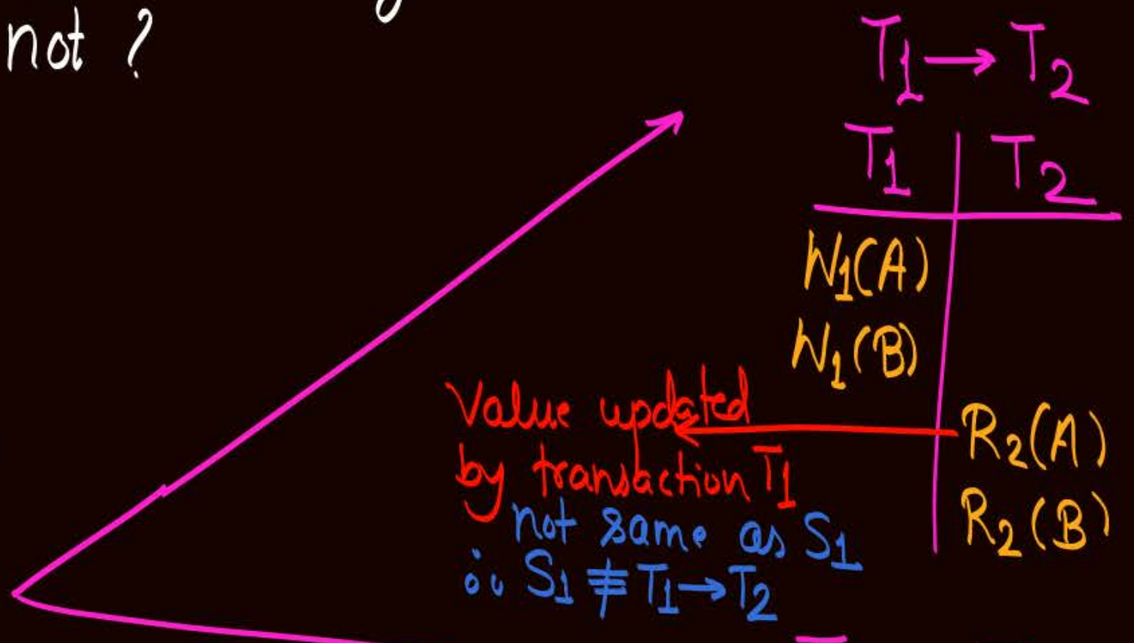


Topic : Serializable schedule

If behaviour of given schedule is equivalent to at least one of the serial schedule (based on the transactions of the given schedule) then. schedule is called serializable schedule

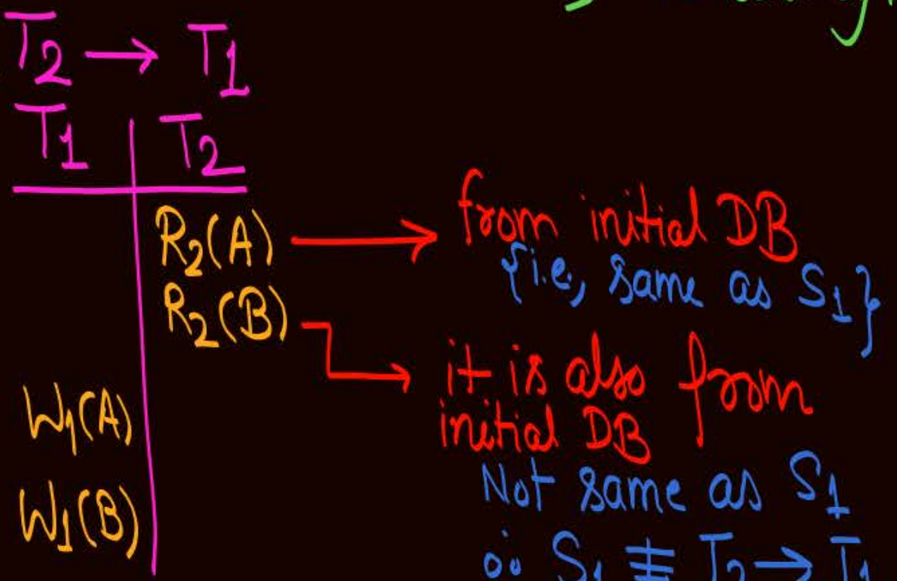
Q: Check whether the given schedule is a serializable schedule or not ?



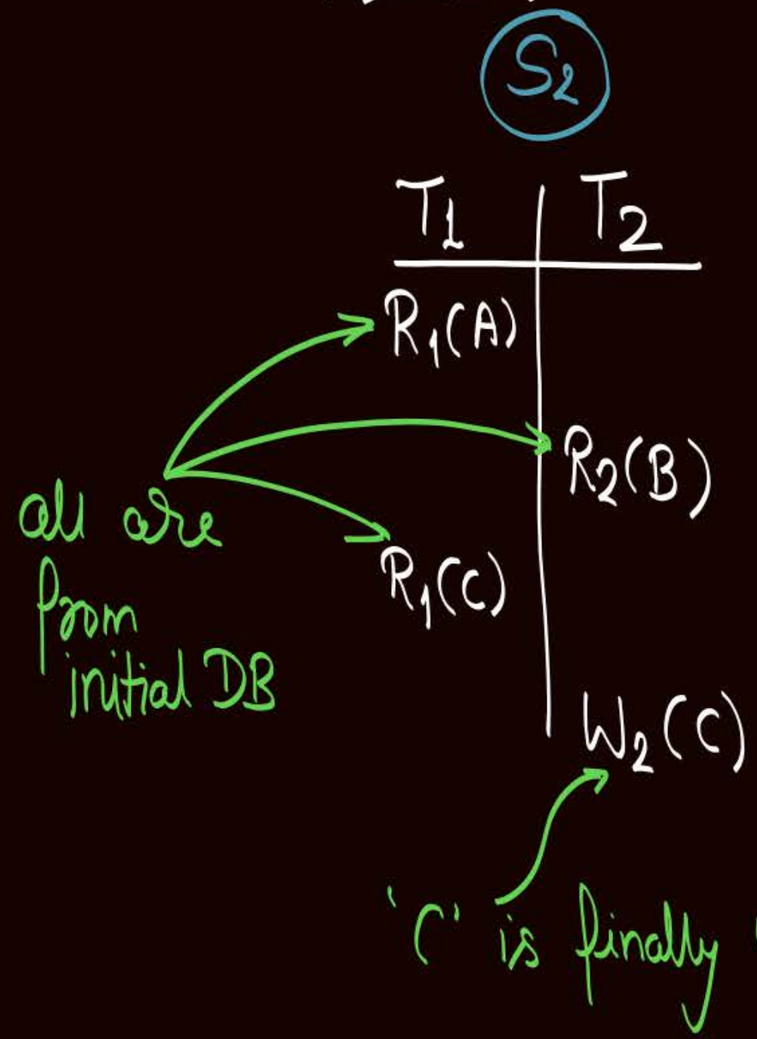


Final write of data item A is by T_1, as well as Final write of data item B is also by T_1

S_1 is not equivalent to any of the serial schedule
∴ S_1 is a non-serializable schedule



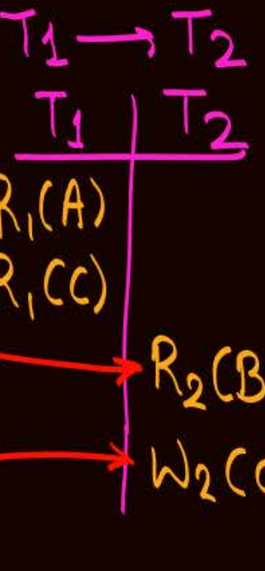
Q: Check whether the given schedule is a serializable schedule or not ?



Same as S_2

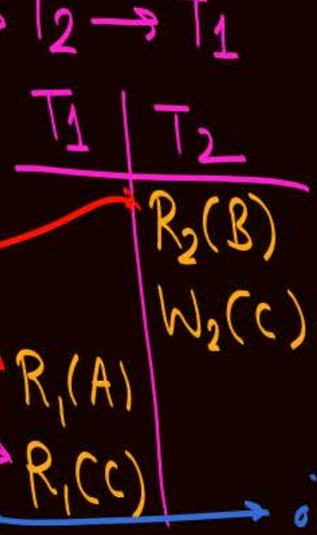
{all from initial DB

{'C' is finally updated by T_2



$S_2 \equiv T_1 \rightarrow T_2$

Schedule S_2 is equivalent to serial schedule $T_1 \rightarrow T_2$ i.e. S_2 is a serializable schedule



from initial DB

$W_2(C)$ ← final update by T_2

Value updated by T_2

Not same as S_2

$\therefore S_2 \neq T_2 \rightarrow T_1$

Q: Check whether the given schedule is a serializable schedule or not ?

S_3

T_1	T_2
$R_1(A)$	
	$R_2(B)$
	$W_2(B)$
$W_1(B)$	

from initial DB

'B' is finally updated by T_1

$T_1 \rightarrow T_2$

T_1	T_2
$R_1(A)$	
$W_1(B)$	
	$R_2(B)$
	$W_2(B)$

'B' is finally updated by T_2 (Not same as S_3)

$\therefore S_3 \not\equiv T_1 \rightarrow T_2$

$T_2 \rightarrow T_1$

T_1	T_2
	$R_2(B)$
	$W_2(B)$
$R_1(A)$	
$W_1(B)$	

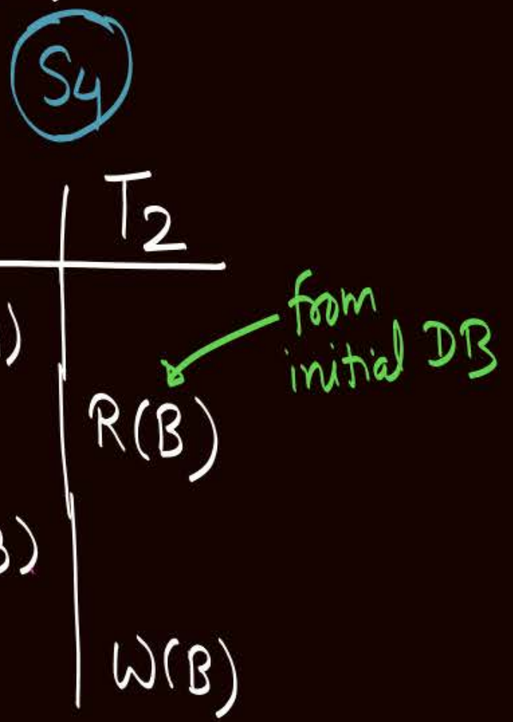
both from initial DB {same as S_3}

'B' is finally updated by T_1 {Same as S_3}

$\therefore S_3 \equiv T_2 \rightarrow T_1$

Schedule S_3 is a serializable schedule, and equivalent serial schedule is $T_2 \rightarrow T_1$

Q: Check whether the given schedule is a serializable schedule or not?



$S_4 \not\equiv T_1 \rightarrow T_2$ because of R(B) of transaction T_2

$S_4 \not\equiv T_2 \rightarrow T_1$ because of final update of data item B

$\Rightarrow$ S_4 is a non-serializable schedule

$T_1 \rightarrow T_2$

T_1	T_2
R(A)	
W(B)	
	R(B)
	W(B)

this R(B) is value updated by T_1



Topic : Problems because of concurrent execution

of transactions

(1) R W Problem / Read-Write Problem / Write after Read Problem

(2) W R Problem / Write-Read problem / Read after write Problem

(3) W W Problem / Write-write Problem / Write after Write problem

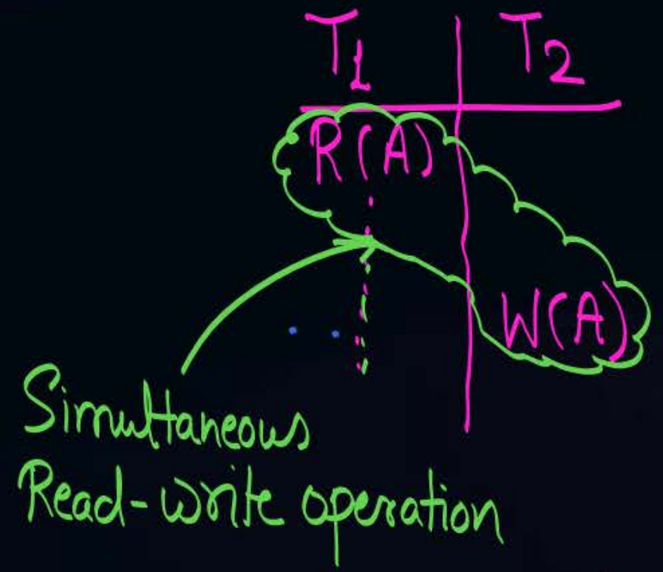
] Exist only if schedule is a non-serializable schedule

(4) Lost update problem] It may exist in non-serializable schedule as well as in serializable schedule

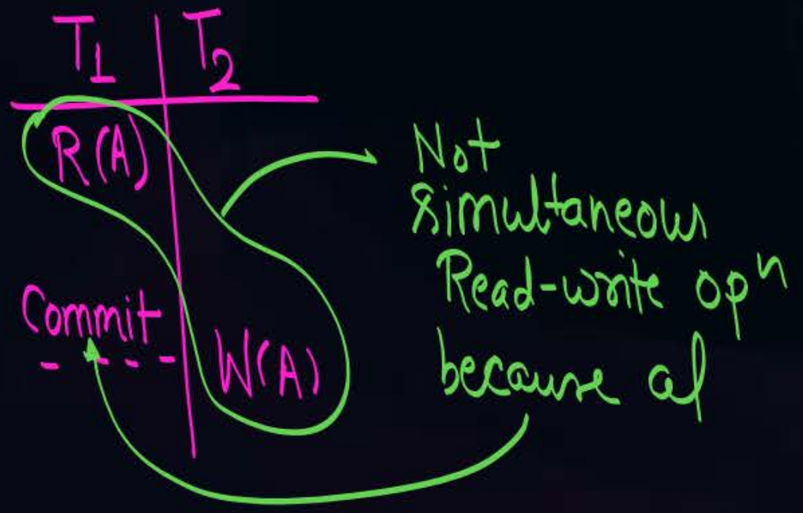


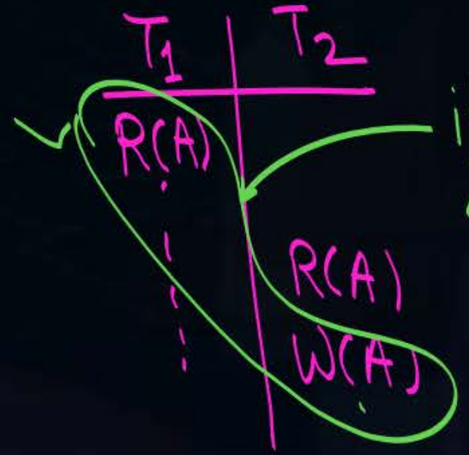
Topic : RW problem

Write after Read Problem :-



If transaction T_2 updates the dataitem A which is already read by an uncommitted transaction T_1, then sequence of operation is called Simultaneous Read-write operation







Topic : RW problem

Write after Read Problem :-

* A schedule is said to have Read-write problem if and only if

(i) Simultaneous Read-Write opⁿ should exist

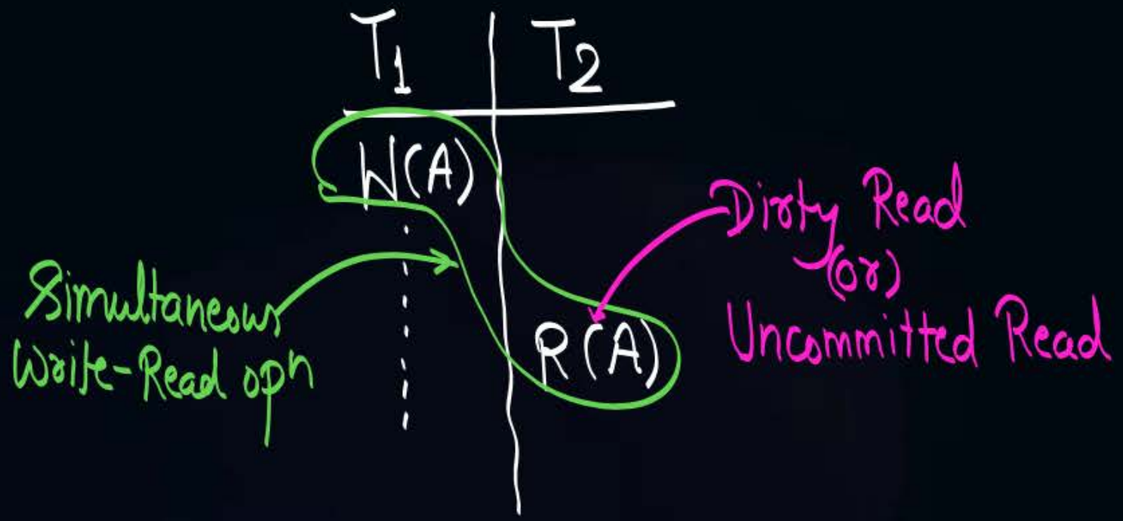
and (ii) Schedule must be a non-serializable schedule



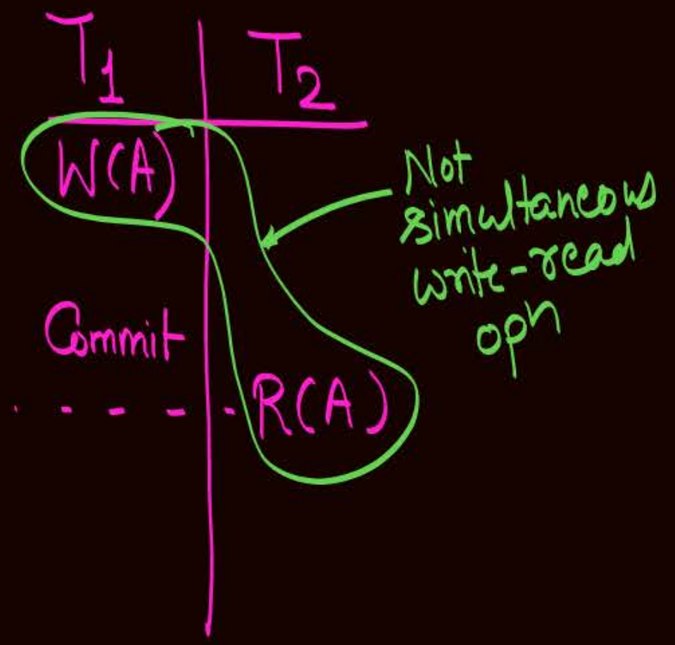
Topic : WR problem

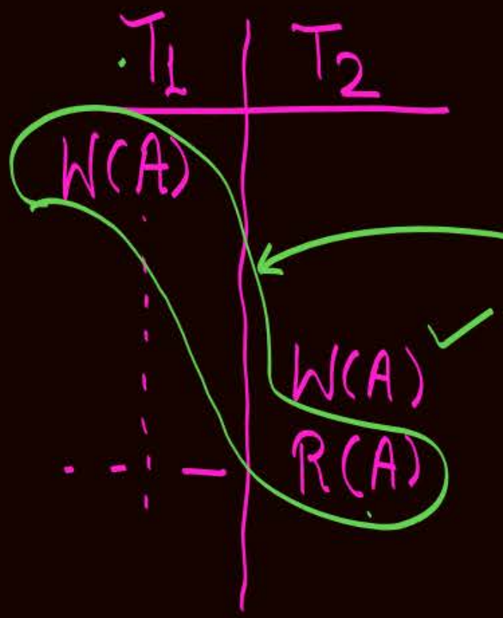
/ Write-Read problem / Read-after-write problem

/ Dirty Read problem / Uncommitted Read problem

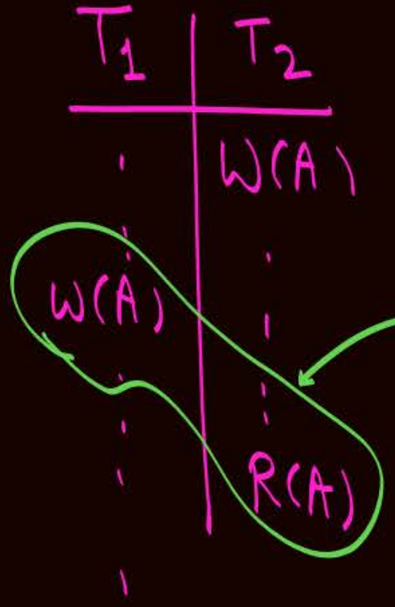


If transaction T_2 reads the value updated by an uncommitted transaction T_1, then sequence of opn is called Simultaneous write-read opn, or Dirty read opn or Uncommitted Read opn





Not a simultaneous write-read opn, because T_2 is reading the value updated by itself. it is not reading the value updated by some other transaction



it is a simultaneous write-read opn



Topic : WR problem

A schedule is said to have Write-read problem if and only if

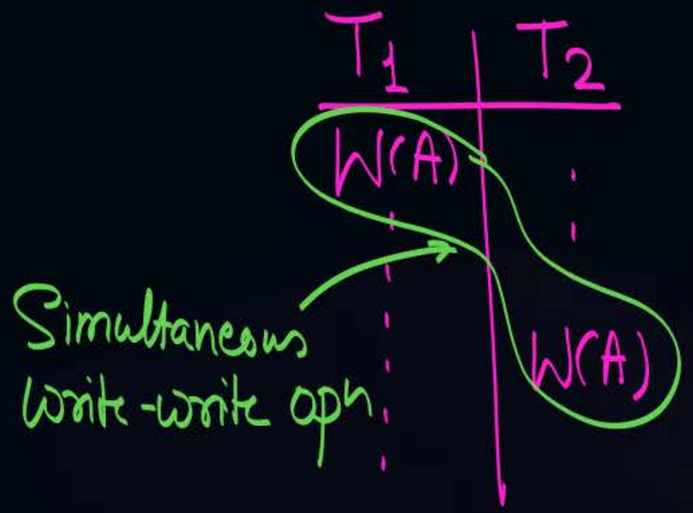
(i) Simultaneous Write-read operation should exist

and (ii) Schedule must be a non-serializable schedule.

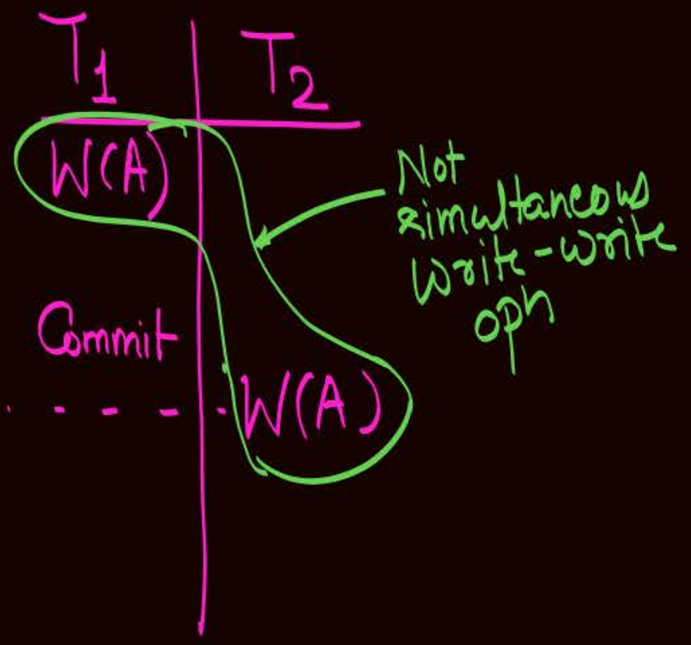


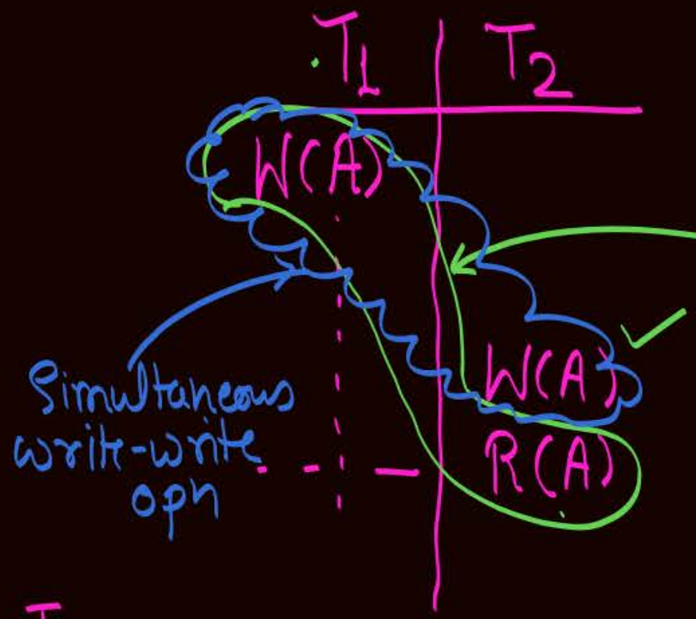
Topic : WW problem

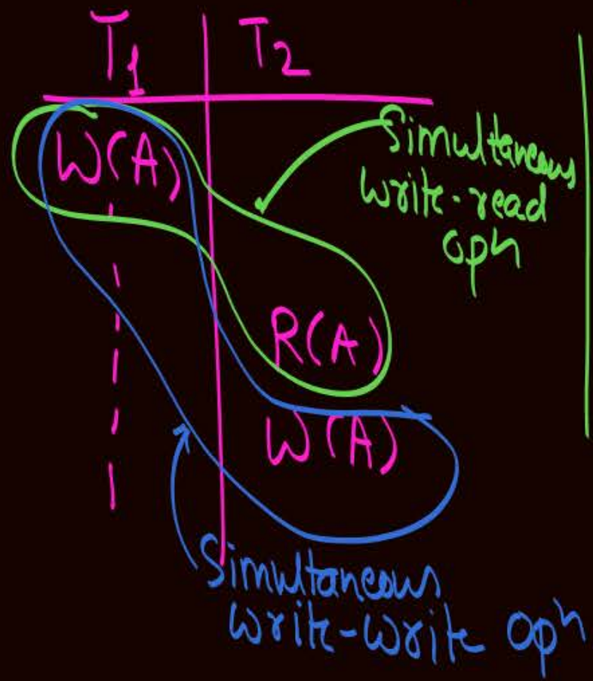
/ Write-Write Problem / Write after Write Problem

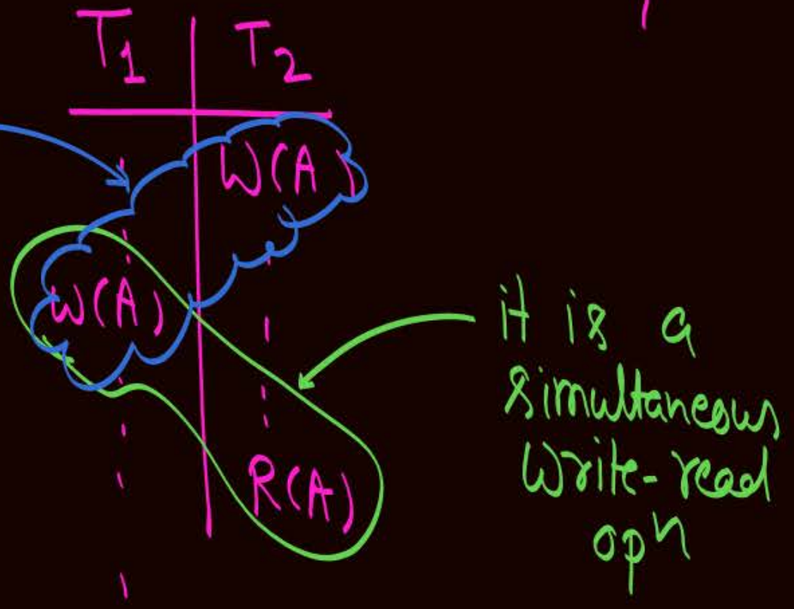


If transaction T_2 updates the data item A already updated by an uncommitted transaction T_1 then sequence of oph is called Simultaneous Write Write oph.


T1
T2
W(A)
Commit
W(A)
Not simultaneous write-write oph


T1
T2
W(A)
Simultaneous write-write oph
W(A)
R(A)
Not a simultaneous write-read oph, because T2 is reading the value updated by itself. it is not reading the value updated by some other transaction


T1
T2
W(A)
Simultaneous write-read oph
R(A)
W(A)
Simultaneous write-write oph


T1
T2
Simultaneous write-write oph
W(A)
W(A)
R(A)
it is a simultaneous write-read oph



Topic : WW problem

/ Write-Write Problem / Write after Write Problem

A schedule is said to have Write-write problem if and only if,

(i) Simultaneous write-write opⁿ should exist,

and (ii) Schedule must be a non-serializable schedule

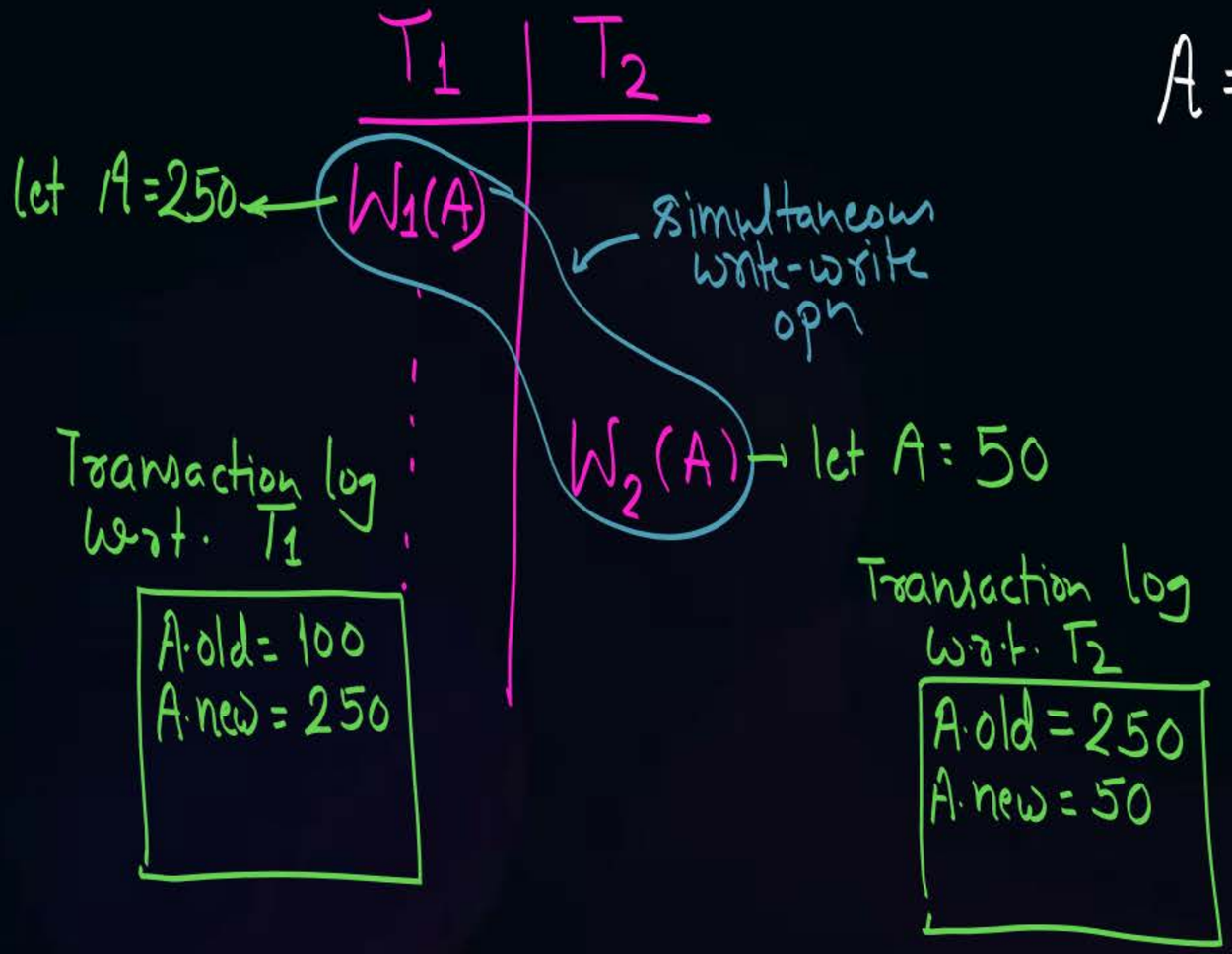


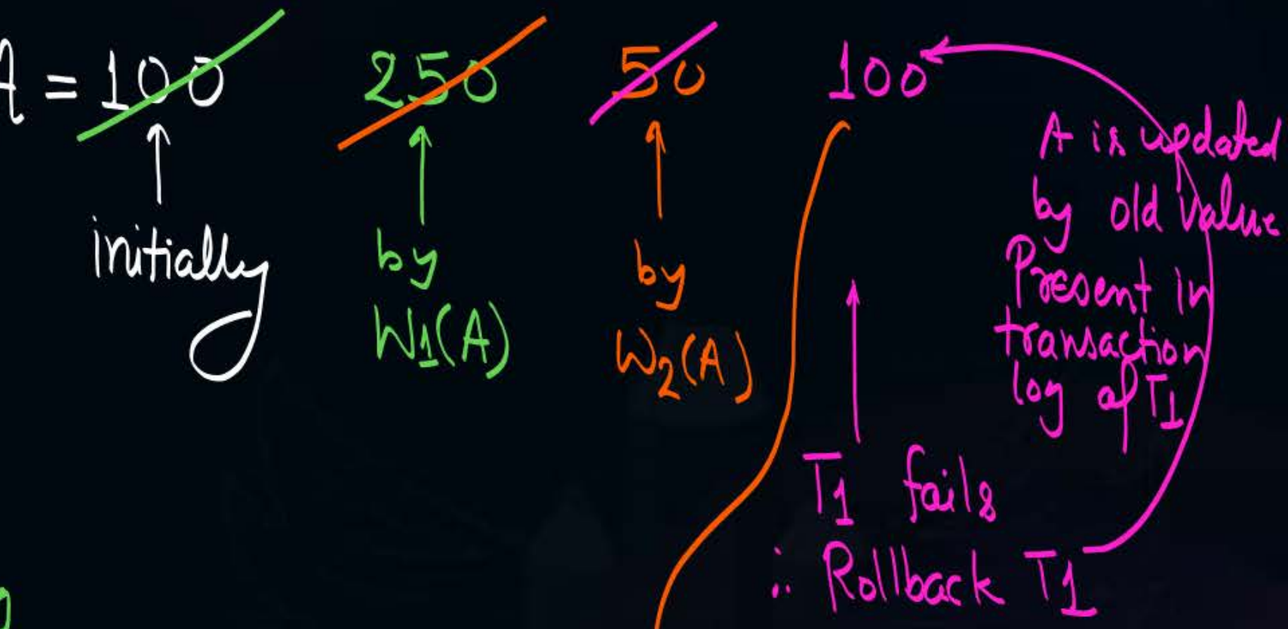
Topic : Lost update problem

Lost update problem exist in a schedule if simultaneous write - write oph exist in the schedule, does not matter whether the schedule is a serializable schedule or a non - serializable schedule

Topic : Lost update problem

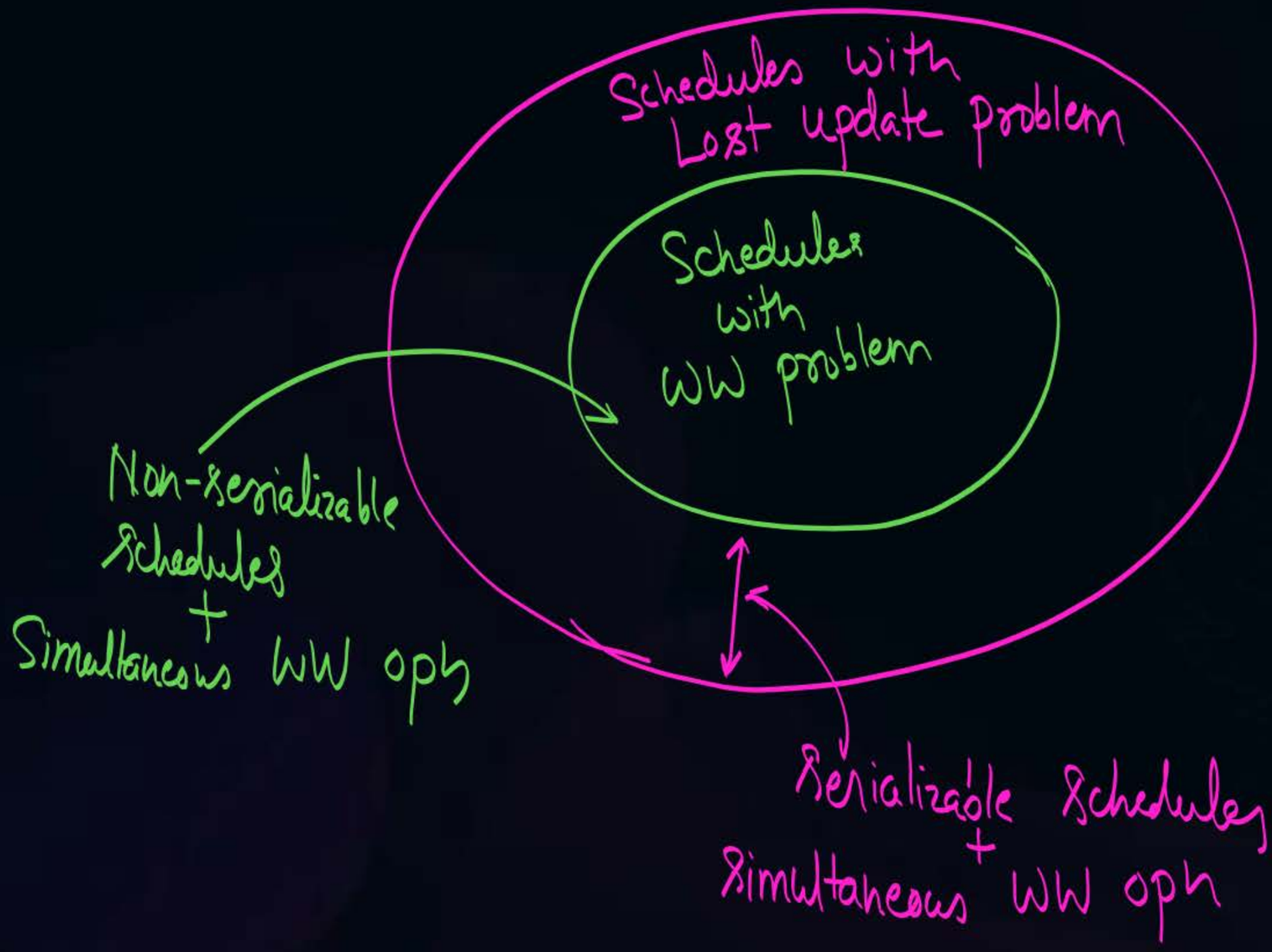
let initial value of A in the database = 100







Topic : Lost update problem



- Every schedule that have WW problem will also have lost update problem, but a schedule with lost update problem may or may not have WW-problem.



2 mins Summary

- Topic: Equivalent schedule
- Topic: Serializable schedule
- Topic: RW problem
- Topic: WR problem
- Topic: WW problem
- Topic: Lost update problem

Slide

THANK - YOU